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Predicting NYC Yellow Taxi Tipping Behavior

A Data-Driven Analysis on Tipping Behavior in NYC 2022

Objective

- Identify key factors influencing tipping behavior
- Analyze when and where tips are higher
- Build a regression model to predict tip amount
- Build an interactive dashboard with NLP integration to show findings

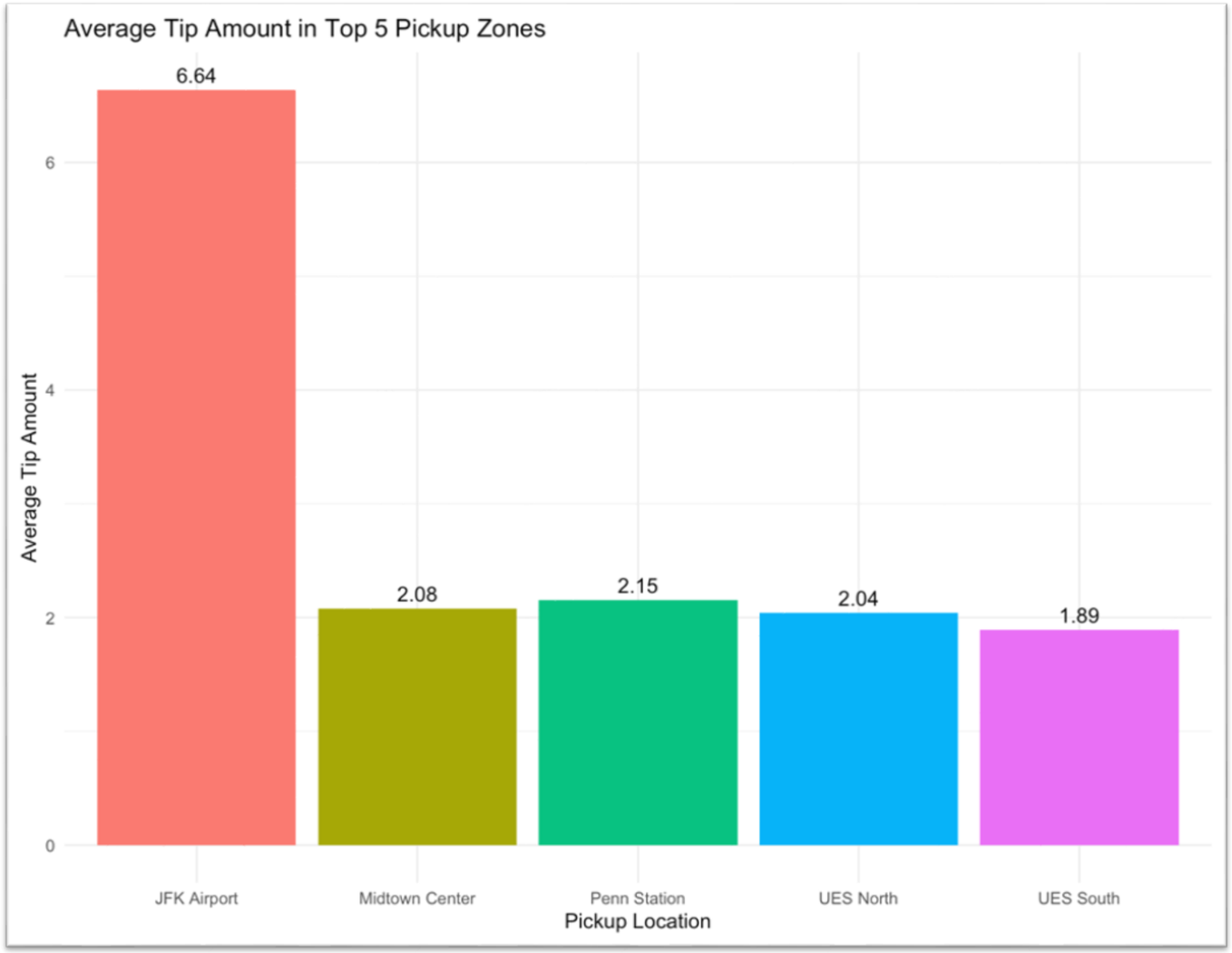
Data Overview

- Dataset: NYC Yellow Taxi Data (January 2022)

- 2+ million trips
  - Random sample of 250k

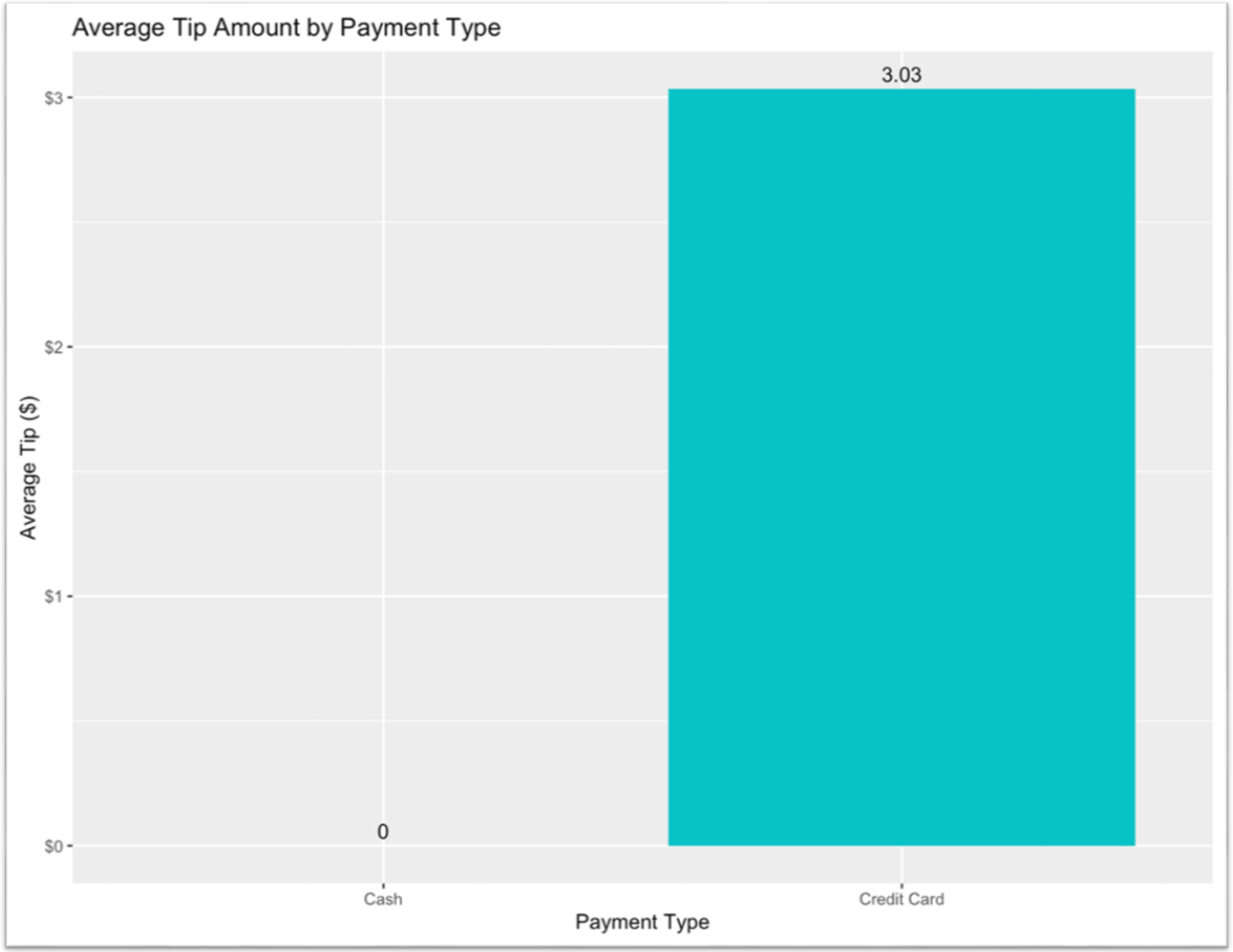
INSIGHT #1: Tipping varies by pickup location

Zone 132 (**JFK Airport**) had the overall highest average tip amount →



INSIGHT #2: Payment method influences tipping

Those who pay with credit cards tip more significantly than those who pay in cash →



Possible Explanations:

Credit card systems make tipping more visible

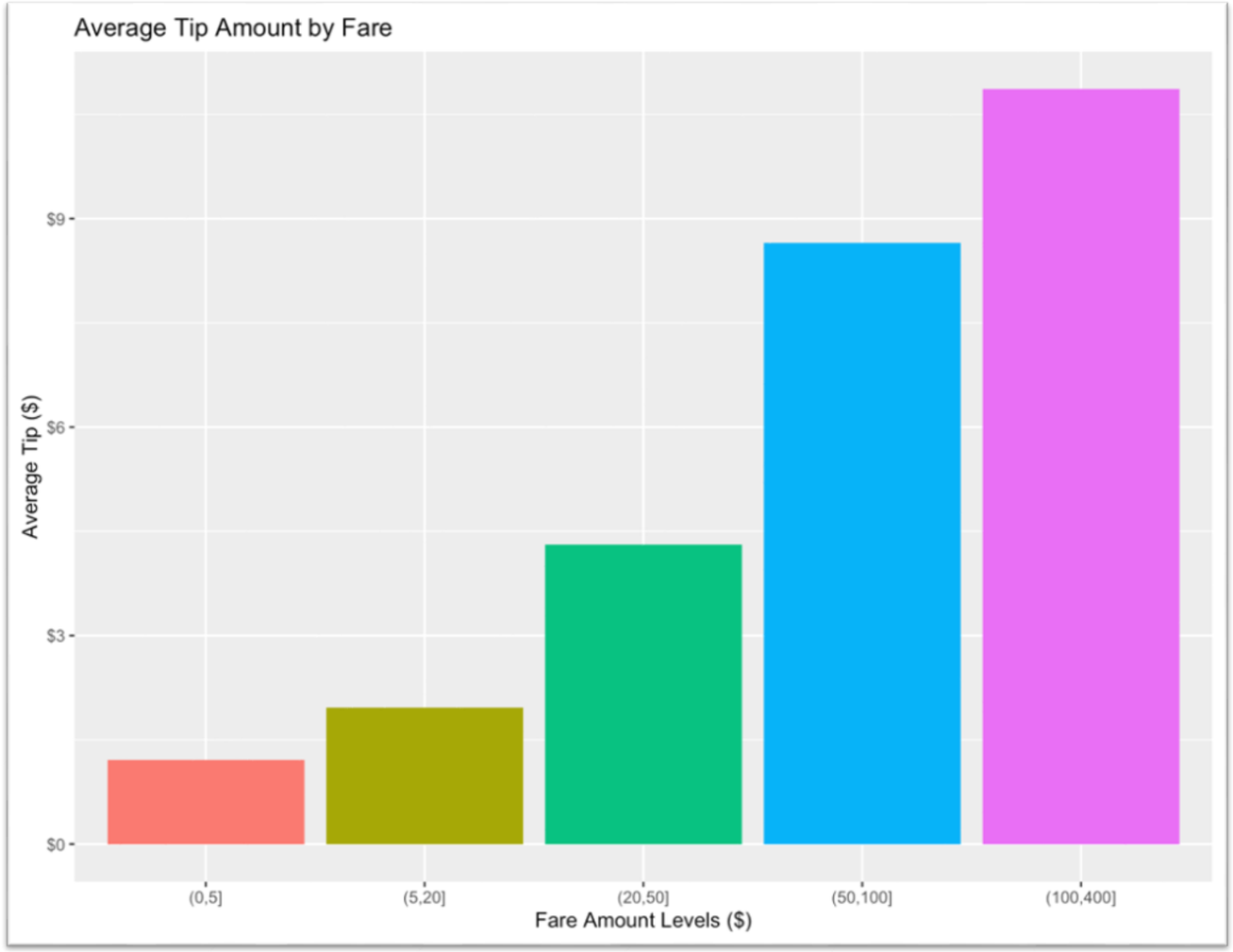
Cash tips are underreported

INSIGHT #3: Higher fares are associated with higher tips

Tip amount increases consistently with fare amount →

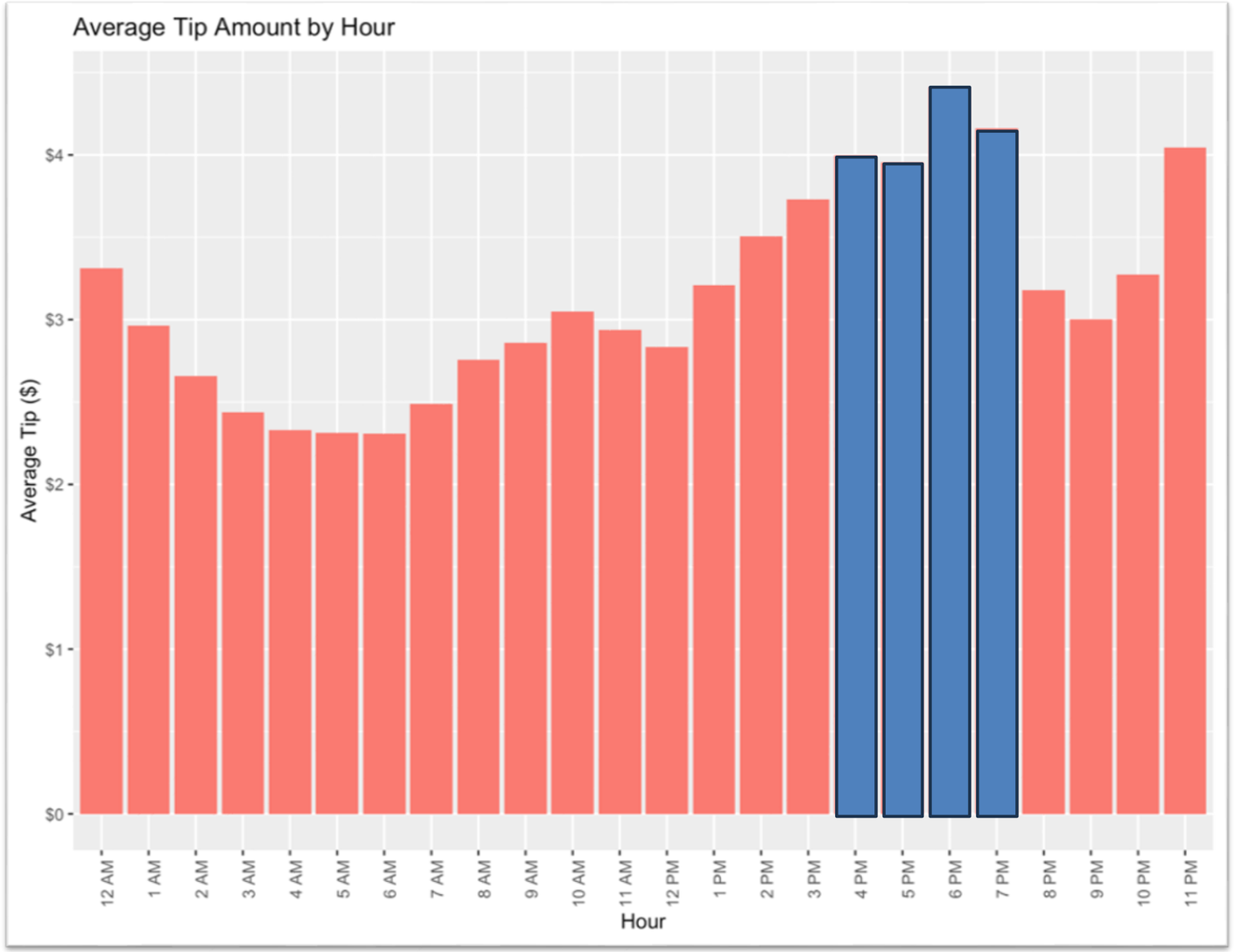
Longer trips are also associated with higher fares

Passengers tip based on cost and trip length



INSIGHT #4: Tipping peaks during the early evening

Tips are generally higher between 4 and 7pm →



Model Overview

- Multiple Linear Regression Model
  - Response: Tip amount
  - Predictors: Fare amount, trip duration, payment type, pickup zone, and pickup hour
- Why these variables
  - Fare amount and trip duration reflect cost
  - Payment type showed clear differences in tipping
  - Pickup zone and hour reflect geographical and temporal factors
- Log transformation applied to tip, fare and duration
  - Improved linearity and reduced skewness in residuals

Model Performance

MAE ≈ 0.85

Predictions are off by approximately $0.85 per trip.

RMSE ≈ 1.89

Most predictions are accurate, **but** a small number of trips may create larger errors.

Overall, this model performs well on average but occasionally makes larger prediction errors.

Addressing Assumptions

Linearity:

- Relationship between predictors and response is not perfectly linear, even after a log transformation

Homoscedasticity:

- Some heteroscedasticity remains
- Variability increases for certain trip types

Normality:

- Residuals are approximately normal
- Heavy tails suggest sensitivity to extreme trips

Independence

- Each trip is treated as an independent observation

These violations suggest that tipping behavior data is complex, but they do not invalidate the usefulness of this model.

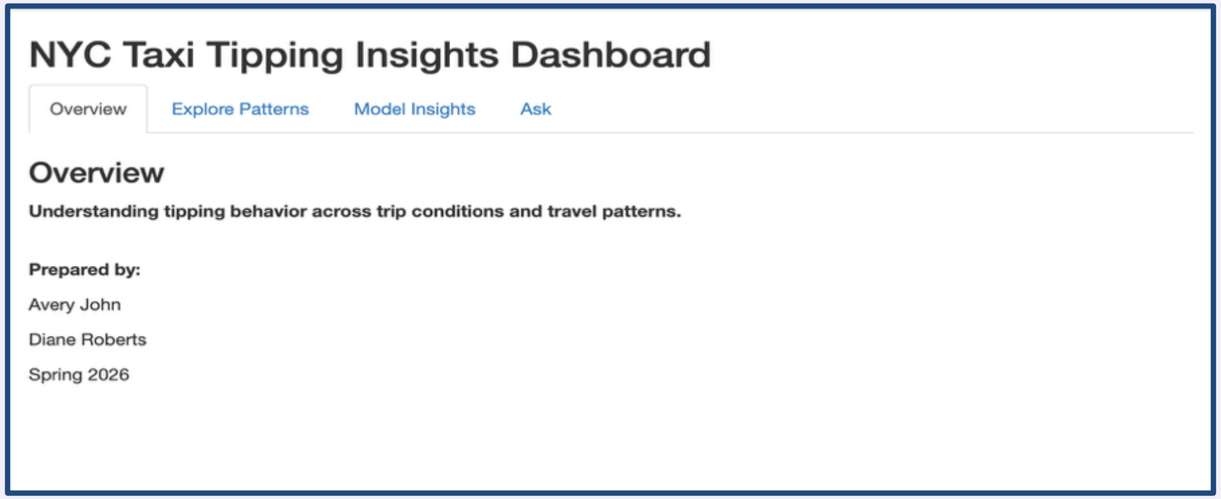
Why This Model was Chosen

- Log transformations reduced skewness and improved model stability
- Includes influential economic and behavioral variables
- Model effectively captures overall trends despite complexity of real-world data

R-SHINY APP

From Insights to Interactive Decision-Making

- Built an interactive R Shiny dashboard
- Enables real-time exploration of tipping patterns
- Integrates predictive modeling + user inputs



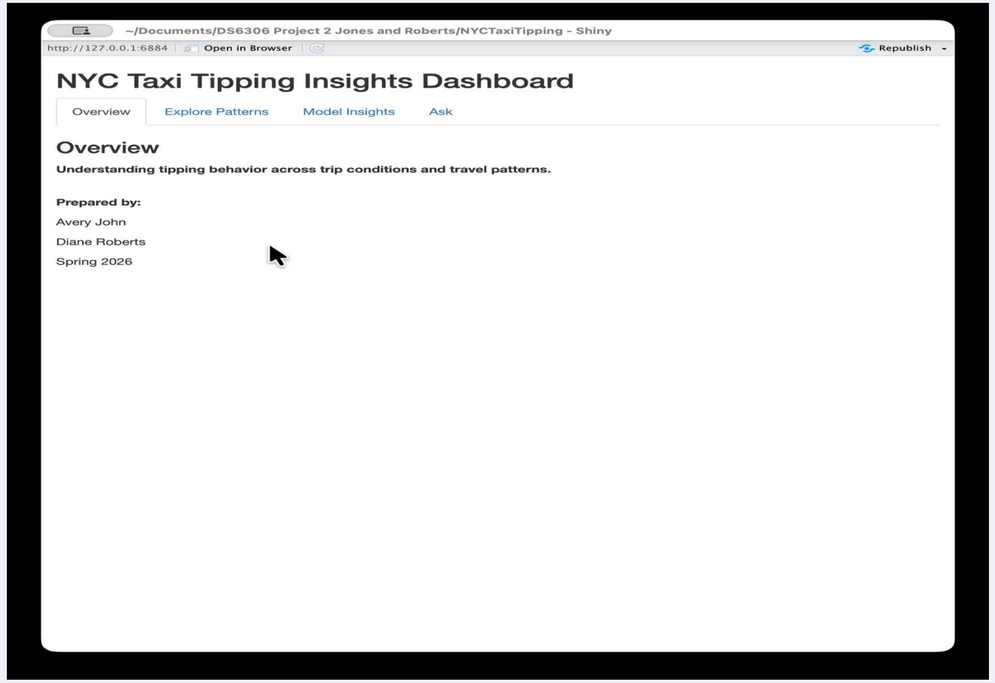
To run the app: ***https://dianer.shinyapps.io/NYCTaxiTipping/***

Exploring Tipping Behavior in Real Time

Compare tipping across:
- Distance groups
- Payment type
- Rate codes

Boxplots show:
- Median (line)
- Mean (red dot)
- Cleaned for interpretability (outliers handled)

Callouts:
- Distance groups ordered logically
- Payment type labeled (Credit Card vs Cash)



Predicting Tip Amounts

User inputs:

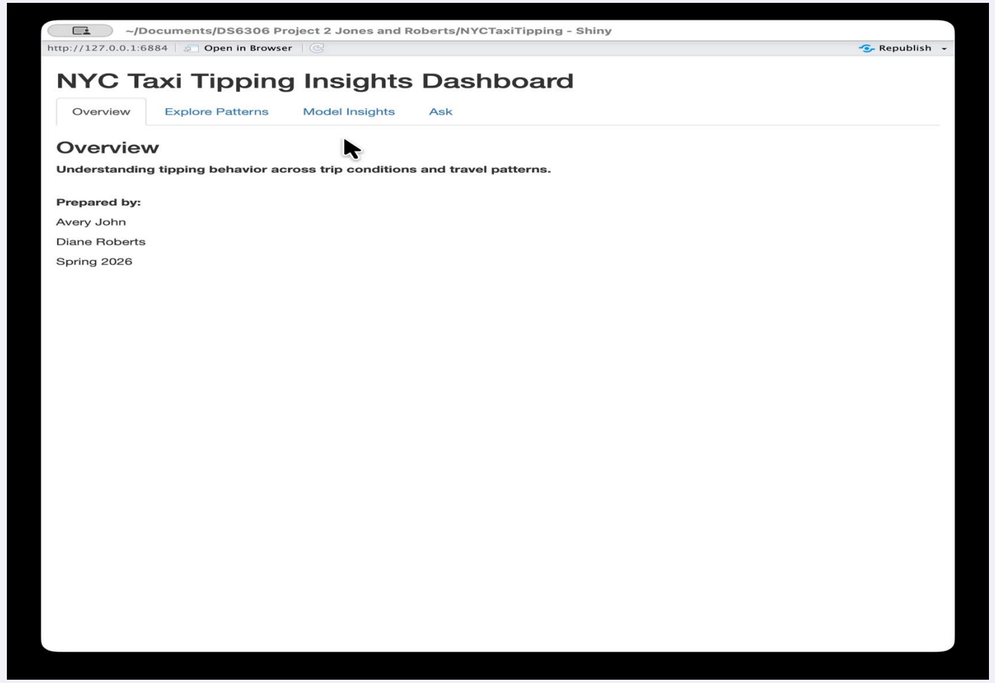
- Fare
- Duration
- Payment type
- Location
- Time of day

Outputs:

- Predicted tip amount
- What drives tip amount

Based on log-linear regression model

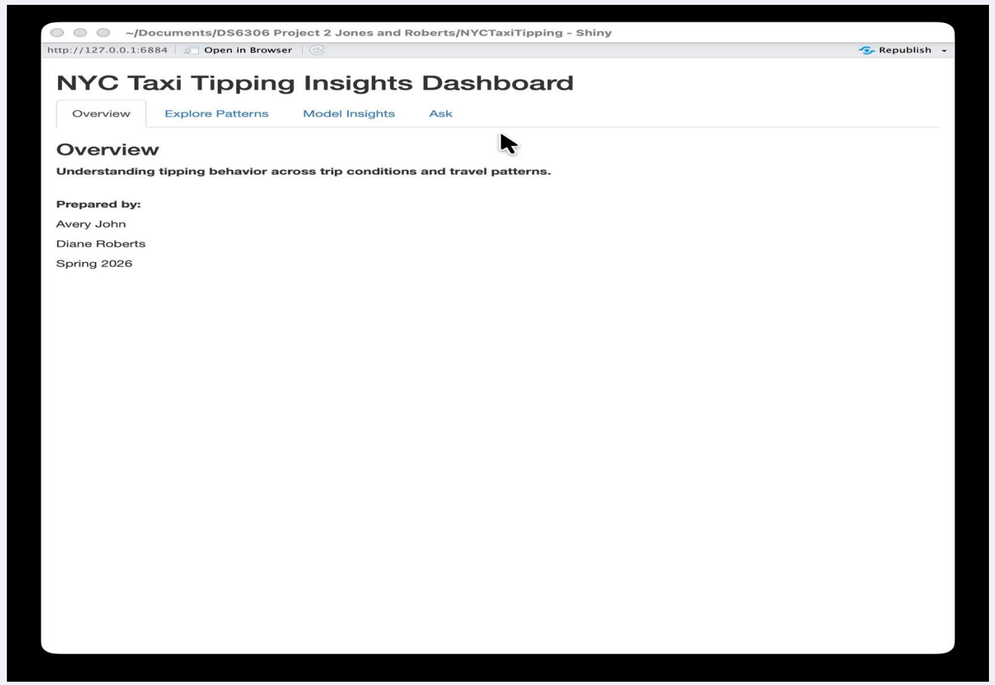
RMSE ≈ 1.89, MAE ≈ 0.85



Adding intelligence with Retrieval-Augmented Generation

RAG proof-of-concept:

- Users can ask natural language questions
- System retrieves relevant insights from project knowledge base
- Not generative AI guessing -> grounded in project findings



QUESTIONS:

- What is the difference between tip amount and tip percentage?
- Why do credit card trips look like they tip more?
- Do airport trips tip differently?
- Do long trips always tip more?
- Why did we use log transformation?
- Why doesn’t the coefficient table change?
- What is this dashboard supposed to do?
- How accurate is the model at predicting tip amounts?

From Analysis to Decision Support

Transforms static analysis into:

- Interactive exploration
- Predictive decision-making
- Explainable insights

Combines:

- Data Science
- UX
- AI-assisted interpretation

Key Takeaways

- **Fare** is the strongest predictor for tipping behavior
  - Other factors like time, location and payment method do also matter
- Final predictive model achieves **strong predictive accuracy** according to MAE and RMSE values

THANK YOU!

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